

More messy diagrammatics — the n -rail challenge

Reply to Gunnar’s note of 7 May 2026

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One-line takeaway

The n -rail count is the matrix element $(L^m)_{00}$ of the **weighted graph Laplacian** $L = D - A$ on K_n with edge weights v_{ij} . For uniform coupling $v_{ij} \equiv v$ this collapses to $W_m^{(n)} = \frac{n-1}{n} (nv)^m$ which extends your $\frac{1}{2}(2v_1)^m$ for $n=2$ to $\frac{2}{3}(3v)^m$, $\frac{3}{4}(4v)^m, \dots$. The loop integrals factor cleanly off the combinatorics: each off-path rail decomposes into a sequence of return-excursions, and each excursion is a single one-loop kinematic kernel.

1 The challenge, restated

You consider a theory with two rail-coupling vertices,

$$\Gamma_{ij}^+ = \begin{array}{c} 1 \\ \text{---} \\ 0 \end{array} \begin{array}{c} 1 \\ \text{---} \\ 0 \end{array} = +v_{ij}, \quad \Gamma_{ij}^- = \begin{array}{c} 1 \\ \text{---} \end{array} \begin{array}{c} 0 \\ \text{---} \end{array} = -v_{ij},$$

with $v_{ij} = v_{ji}$ the symmetric coupling between rail i and rail j , and the propagator on every rail is $G(\omega) = (-i\omega + r)^{-1}$.

You restrict to n -rail diagrams with one rail of type 1 (the “1-rail”—initially rail 0) and $n - 1$ rails of type 0. By tracking “the path the 1 takes,” you arrive at the local rule:

- every vertex where the 1 *stays* on its current rail and a bridge crosses to another rail j contributes $+v_{aj}$;
- every vertex where the 1 *swaps* from rail a to rail b contributes $-v_{ab}$.

For $n = 2$ the m -th-order weighted sum is $\frac{1}{2}(2v_1)^m$, and the open question is

The two challenges

- (C1) Extend $\frac{1}{2}(2v_1)^m$ to $n = 3, 4, \dots$ in closed form, with the v_{ij} kept symbolic (so that one can read off which couplings contribute at which order).
- (C2) Account for the loop integrals (the analytic side — not just the symbol-level sign and weight) within the AC framework.

2 The combinatorial answer (C1): walks on the weighted complete graph

The transfer step. Between two consecutive vertices on the 1’s path, the 1 sits on a single rail a . At the next vertex it either

- *stays* on rail a , with a bridge to some rail $j \neq a$, weight $+v_{aj}$;
- *swaps* from rail a to a chosen rail $b \neq a$, weight $-v_{ab}$.

Define the rail-to-rail transfer matrix T with entries

$$T_{aa} = \sum_{j \neq a} v_{aj} \quad (\text{stay, summed over bridge target}), \quad T_{ab} = -v_{ab} \quad (a \neq b, \text{ swap}).$$

Read off the structure: T_{aa} is the weighted degree of vertex a in the complete graph K_n with edge weights v_{ij} , and T_{ab} is the *negated* adjacency. Therefore

The transfer matrix is the graph Laplacian

$$T = L = D - A, \quad D_{aa} = \sum_{j \neq a} v_{aj}, \quad A_{ab} = v_{ab} (1 - \delta_{ab}).$$

The whole sign-and-symmetry-factor accounting collapses into one line: the 1's path is a length- m walk on $\{0, 1, \dots, n-1\}$ generated by L , beginning and ending at rail 0, so the order- m weighted sum is

$$W_m^{(n)} = (L^m)_{00}.$$

Sanity check: $n = 2$. With one coupling $v_{01} \equiv v$ and rails $\{0, 1\}$,

$$L = v \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad \text{eigenvalues } \{0, 2v\}, \quad \text{eigenvectors } \frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(1, -1).$$

Then $L^m = (2v)^m P_{\perp}$ with $P_{\perp} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, and $(L^m)_{00} = \frac{1}{2}(2v)^m$ — exactly your formula. ✓

General n , uniform coupling. If $v_{ij} \equiv v$, then $L = v(nI - J)$ where J is the all-ones matrix. The spectrum is $\{0, nv, nv, \dots, nv\}$ (multiplicity $n-1$ on the $\mathbf{1}^{\perp}$ subspace), so

$$L^m = (nv)^m P_{\perp}, \quad P_{\perp} = I - \frac{1}{n} J,$$

and the boundary matrix element is

$$W_m^{(n)} = (L^m)_{00} = \frac{n-1}{n} (nv)^m. \quad (1)$$

This is the requested extension. The first values are

n	2	3	4	5
$W_m^{(n)}$	$\frac{1}{2}(2v)^m$	$\frac{2}{3}(3v)^m$	$\frac{3}{4}(4v)^m$	$\frac{4}{5}(5v)^m$

The pattern is intuitive: at order m there are n^m raw step sequences (at each of m vertices the 1 chooses one of $n-1$ partner rails, with two moves per partner — stay or swap, signed); the projection onto walks returning to rail 0 extracts the $\frac{n-1}{n}$ Frobenius factor.

General n , distinct v_{ij} . Without uniformity the answer is the spectral expansion (for $m \geq 1$)

$$W_m^{(n)} = \sum_{\alpha=1}^{n-1} \lambda_{\alpha}^m |\psi_{\alpha}(0)|^2, \quad (2)$$

with $L\psi_{\alpha} = \lambda_{\alpha}\psi_{\alpha}$ orthonormal and $\lambda_0 = 0$ dropping out for any $m \geq 1$ (the constant eigenvector $\psi_0 \propto \mathbf{1}$ sits in the kernel of L). For the symbolic v_{ij} , this is

$$W_m^{(n)} = \mathbf{e}_0^{\top} L^m \mathbf{e}_0, \quad L_{ij} = \delta_{ij} \sum_{k \neq i} v_{ik} - (1 - \delta_{ij}) v_{ij}.$$

For example at $n = 3$ in the basis $\{0, 1, 2\}$,

$$L = \begin{pmatrix} v_{01} + v_{02} & -v_{01} & -v_{02} \\ -v_{01} & v_{01} + v_{12} & -v_{12} \\ -v_{02} & -v_{12} & v_{02} + v_{12} \end{pmatrix},$$

and $(L^2)_{00} = (v_{01} + v_{02})^2 + v_{01}^2 + v_{02}^2$, $(L^3)_{00} = (v_{01} + v_{02})^3 + v_{01}^2(v_{01} + v_{02}) + v_{02}^2(v_{01} + v_{02}) + v_{01}^2(v_{01} + v_{12}) + v_{02}^2(v_{02} + v_{12}) + 2v_{01}v_{02}v_{12}$, which one can read term-by-term as the path enumeration on the rail graph (“stay-stay”, “stay-swap-swap”, etc.).

Connection to the bidegree-(1,1) Q1 result. This is the n -rail generalisation of the $n=2$ ladder result we sent on 29 Apr: there, projecting the four-variable vertex polynomial $\hat{V}$ onto the bidegree-(1,1) sector gave $\hat{T}(z, w) = (z_1 - z_2)(w_1 - w_2)$, eigenvalue $\lambda=2$, and $\langle b | \hat{T}^n | b \rangle = 2^n$. The Laplacian above is exactly that boundary-projected operator written in rail-state basis: the antisymmetric mode of K_2 is the $\mathbf{1}^\perp$ eigenspace of L . For $n \geq 3$ the antisymmetric/symmetric language disappears and we need the full Laplacian spectrum — but the AC bookkeeping is identical.

3 The analytic answer (C2): loop integrals attach as kinematic atoms

The combinatorial sum (1)–(2) only counts *symbol-level* weights. The analytic value of one diagram — the loop integrals over internal frequencies / momenta — is built on top of that count. The key observation is that for a fixed rail-walk w , the diagram value factorises into two independent pieces:

The recipe for one diagram

$$\Gamma(w) = \underbrace{S(w)}_{\text{symbol weight}} \times \underbrace{K(w)}_{\text{kinematic kernel}}$$

where $S(w)$ is a product of vertex weights $\pm v_{ij}$ (no integrals) and $K(w)$ is a product of propagator chains and loop integrals (no v_{ij}). They never mix.

The full m -th order contribution to the impurity self-energy is then $\Gamma^{(m)} = \sum_w S(w)K(w)$ summed over all length- m walks $0 \rightarrow 0$. The Laplacian formula $\sum_w S(w) = (L^m)_{00}$ from §2 collects the symbol part; what is left is to write down $K(w)$.

Step 1: how to read off the symbol weight $S(w)$

A walk $w = (\text{vertex}_1, \text{vertex}_2, \dots, \text{vertex}_m)$ is a sequence of m moves, each one labelled by either “stay” or “swap” and the rail j involved. The recipe is local — one factor per vertex — and just transcribes Gunnar’s local rule:

$$S(w) = \prod_{t=1}^m \sigma_t v_{a_t j_t}, \quad \sigma_t = \begin{cases} +1 & \text{if vertex } t \text{ is a stay} \\ -1 & \text{if vertex } t \text{ is a swap} \end{cases}, \quad (3)$$

where a_t is the rail the 1 sits on just before vertex t and j_t is the partner rail at that vertex. *Example:* the walk “stay-bridge-to-1, swap-onto-2, swap-back-to-0” on rails $\{0, 1, 2\}$ gives $S = (+v_{01})(-v_{02})(-v_{02}) = +v_{01}v_{02}^2$. This is just the product Gunnar wrote down; nothing analytic happens here.

Step 2: how to compute the kinematic kernel $K(w)$

The kernel is built from the propagator structure of the diagram. Two ingredients, in order:

(a) The on-path piece (chain on the 1's worldline). Read the 1's worldline left-to-right. Between consecutive vertices the 1 sits on some rail and carries the external frequency ω , contributing one propagator $G(\omega) = (-i\omega + r)^{-1}$. With m vertices there are $m+1$ such segments, so

$$K_{\text{on-path}}(w) = \prod_{\text{segments}} G(\omega) = \frac{1}{(-i\omega + r)^{m+1}}. \quad (4)$$

This factor is the same for every walk of length m — it does not care about chunk pattern — and is just the propagator dressing of the external 1-line.

(b) The off-path piece (one loop per visited bath rail). Now consider each bath rail j *separately*. Count how many “stay-with-bridge-to- j ” vertices appear in w ; call this number $p_j(w)$. Then rail j has p_j insertions and (because it has no external sources) closes into a single **p_j -cycle loop integral**:

$$K_j(p_j) := \int \frac{d\omega'}{2\pi} \prod_{k=1}^{p_j} G_j(\omega' - \Omega_k^{(j)}), \quad (5)$$

where the $\Omega_k^{(j)}$ are external-frequency injections at the j -rail vertices (determined by where in the walk the rail- j visits happen). The full kernel is then

$$K(w) = K_{\text{on-path}}(w) \cdot \prod_{j: p_j \geq 1} K_j(p_j). \quad (6)$$

The standard names of the $K_j(p_j)$ are familiar:

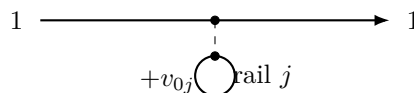
p_j	name	$K_j(p_j)$
1	tadpole	$\int d\omega' G_j(\omega')$
2	bubble	$\int d\omega' G_j(\omega') G_j(\omega' - \Omega)$
3	sunset (2-loop)	$\int d\omega'_1 d\omega'_2 G_j G_j G_j$
p	$(p-1)$ -loop p -cycle	$\int \prod_k G_j(\omega' - \Omega_k)$

Swap vertices do *not* contribute insertions on bath rails — they live on the 1's worldline (they change which rail the 1 is sitting on, no extra bath-rail visit). So p_j counts only stays with bridge-to- j .

Why the kernel depends only on the chunk pattern. Two walks with the same chunk pattern (same sequence of run-lengths of the 1 on a single rail) have the same multiset $\{p_j\}$ of off-path insertion counts, up to relabelling of bath rails. They therefore share $K(w)$. This is why *two distinct walks can carry the same kernel* — and why the K -library is much smaller than the walk count.

Worked examples (a small atlas). Four canonical low-order diagrams illustrate how the recipe (Steps 1–2 above) attaches each rail-walk to an explicit $S(w) \cdot K(w)$ value. Solid lines are propagators (arrow on the 1-rail to mark the impurity worldline), dashed verticals are 4-point vertices with the indicated $\pm v_{ij}$ weight, and a closing arc on a bath rail indicates the rail closing into a loop. Each box states $S(w)$ from (3), the per-rail insertion counts $\{p_j\}$, and $K(w)$ from (4)–(6).

(i) Order $m=1$, *stay-vertex with bridge* $0 \rightarrow j$. The 1 stays on rail 0; rail j is visited once and closes into a tadpole.

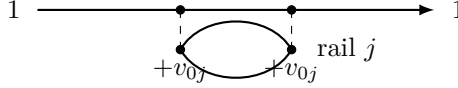


Apply the recipe. Step 1: one stay vertex, $\sigma_1 = +1$, partner rail $j \Rightarrow S = +v_{0j}$. Step 2: $p_j=1$ (single bridge to j), so $K_j = \mathcal{T}$ (tadpole), and the on-path chain has $m+1=2$ propagators:

$$S = +v_{0j}, \quad K = \frac{1}{(-i\omega + r)^2} \cdot \underbrace{\int \frac{d\omega'}{2\pi} \frac{1}{-i\omega' + r}}_{\mathcal{T}(r)}.$$

Chunk pattern: (2) on rail 0. The tadpole is absorbed into mass renormalisation in any sensible scheme.

(ii) Order $m=2$, two stays both bridging to the same rail j . Rail j is visited at two consecutive insertions; the segment between them closes into a bubble.

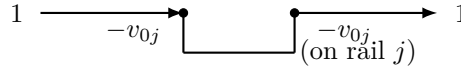


Apply the recipe. Step 1: two stay vertices, both with partner rail $j \Rightarrow S = (+v_{0j})(+v_{0j}) = v_{0j}^2$. Step 2: $p_j=2$ insertions $\Rightarrow K_j = \mathcal{B}$ (bubble), on-path chain $1/(-i\omega + r)^3$:

$$S = v_{0j}^2, \quad K = \frac{1}{(-i\omega + r)^3} \cdot \underbrace{\int \frac{d\omega'}{2\pi} G_j(\omega') G_j(\omega' - \Omega)}_{\mathcal{B}(\Omega; r)}.$$

Chunk pattern: (3) on rail 0. The bubble's external Ω is the frequency flowing through the 1 between the two vertices.

(iii) Order $m=2$, swap-and-return $0 \rightarrow j \rightarrow 0$. The 1 swaps onto rail j at the first vertex and back at the second. There is *no closed loop*: the rail- j segment between the swaps is part of the 1's worldline, not a bath excursion.

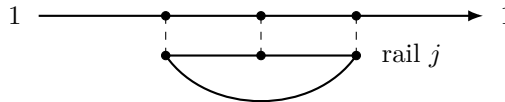


Apply the recipe. Step 1: two swap vertices $\Rightarrow S = (-v_{0j})(-v_{0j}) = +v_{0j}^2$. Step 2: *no* stay-vertex bridges, so $p_j=0$ for every bath rail — **no loop integrals at all**. The whole kernel is just the on-path chain (now passing through three rail-segments: $0 \rightarrow j \rightarrow 0$):

$$S = +v_{0j}^2, \quad K = \frac{1}{(-i\omega + r)^3} \quad (\text{chain only, no } \int d\omega').$$

Chunk pattern: (1, 1, 1) — one segment on rail 0, one on rail j , one back on rail 0. **Compare with (ii):** same symbol weight $+v_{0j}^2$, completely different kernel. The Laplacian $(L^2)_{00}$ collects both into the $2v_{0j}^2$ coefficient at the symbol level; the recipe keeps them distinct on the analytic side. This is the cleanest demonstration that S and K are independent factors.

(iv) Order $m=3$, three stays bridging to the same rail j . Rail j has three consecutive insertions; the j -segments form a closed 3-cycle = sunset (2-loop).



Apply the recipe. Step 1: three stays, all to rail $j \Rightarrow S = (+v_{0j})^3$. Step 2: $p_j=3 \Rightarrow K_j = \mathcal{S}$ (sunset, 2-loop):

$$S = v_{0j}^3, \quad K = \frac{1}{(-i\omega + r)^4} \cdot \underbrace{\int \frac{d\omega'_1 d\omega'_2}{(2\pi)^2} G_j(\omega'_1) G_j(\omega'_2) G_j(\omega'_1 + \omega'_2 - \Omega_{12})}_{\mathcal{S}(\Omega_1, \Omega_2; r)}.$$

Chunk pattern: (4) on rail 0. Standard 2-loop sunset.

At higher order m , partitions of $m+1$ index the kernel library: $(m+1) \rightarrow$ closed cycle on one bath rail (single-rail m -loop), $(m, 1) \rightarrow$ closed cycle plus tadpole, etc. Cases (ii)–(iv) sit in the all-bridges sub-class; case (iii) is representative of the swap-induced no-loop pieces. The rest of the chunk-pattern library combines these.

Why the kernels are bounded in number. This is the *rein-in* you asked about: even though the number of full diagrams grows like $(nv)^m$, the number of *distinct kinematic kernels* is bounded by the partition number $p(m+1)$ ($=1, 2, 3, 5, 7, 11, 15, 22, \dots$): two walks with different rail labels but the same chunk-length pattern (the partition of $m+1$ into the consecutive runs of the 1 on a single rail) share their kernel. The actual count is generically below $p(m+1)$ because not every partition is realised at a given n ; e.g. at $n=3, m=5$ direct enumeration gives 8 distinct chunk-patterns (verified in `poc_laplacian.py`), all 342 raw trajectories collapse onto these 8 master integrals weighted by Laplacian path multiplicities. *This* is what AC buys you on the analytic side: a contraction from (many walks) $\rightarrow$ (a finite kernel library).

Where the AC lens actually kicks in (and where it does not). The recipe in Steps 1–2 is just Gunnar’s local rule transcribed diagram-by-diagram — there is no AC *magic* in it. The AC lens enters when we trade the per-diagram view for the **generating function view**: package the entire perturbation series into one object and let its analytic properties answer questions about the whole series at once.

Concretely, define

$$\mathcal{G}^{(n)}(z; \{v_{ij}\}) := \sum_{m \geq 0} W_m^{(n)} z^m = [(I - zL)^{-1}]_{00},$$

which packages the symbol-side (Laplacian path weights) at all orders. Many questions which *look* like “please compute and tabulate 10^6 diagrams” become one-line readings off $\mathcal{G}^{(n)}(z; \cdot)$:

Translation table: per-diagram question $\iff$ AC reading

Per-diagram question	AC reading on $\mathcal{G}$
“At what rate does $W_m^{(n)}$ grow as $m \rightarrow \infty$?”	Dominant pole of $\mathcal{G}$ at $z = \lambda_1^{-1}$ (F-S Thm. V.7); for uniform v , $\lambda_1 = nv$.
“What is the exact prefactor of $(nv)^m$?”	Residue of $\mathcal{G}$ at the dominant pole; for uniform v , $\frac{n-1}{n}$ from the numerator zero $1-zv$.
“Order- m split by channel 0–1 usage?”	Mark $v_{01} \rightarrow v_{01}u$; coefficient of u^k counts walks with exactly k uses of v_{01} (§4 (3)).
“Forbid one channel: new growth?”	Set $v_{ij}=0$ in $\mathcal{G}$; new dominant pole gives the restricted rate (§4 (3)).
“All-orders foreground given background integrated out?”	Schur reduce $\mathcal{G}$: one rational function with frequency-dependent self-energy $\Sigma(z)$ (§4 (4)).
“Truncate to short off-rail propagation?”	Matrix continued fraction obtained by recursive Schur peel; depth- k truncation keeps $\leq k$ powers of the off-rail block (§4 (5)). Used as a structural device; the precise walk-counting interpretation depends on the truncation convention.

The point: the rules of the diagram game are local (one factor per vertex), but *global* questions about the perturbation series become poles, residues, zeros, and rational-function manipulations of $\mathcal{G}$. The AC lens is precisely the dictionary between the two. §4 develops it into a working toolkit;

the one-line takeaway is that *any time you need to know something about “all the diagrams” without computing them, the right object to manipulate is $\mathcal{G}^{(n)}(z; \{v_{ij}\})$.*

What the AC lens *does not* buy you is the explicit kinematic kernel for an individual chunk pattern — those tadpole / bubble / sunset integrals are still genuine analytic work; AC reduces *how many distinct ones you need* (the $p(m+1)$ bound), not the difficulty of any single one.

Reading off the symmetry factor of each kernel. A natural follow-up: *can we read off the multiplicity / symmetry factor of each kernel from the AC machinery?* Yes, in closed form. For an unordered chunk-length partition $\lambda = (\ell_1, \dots, \ell_k)$ of $m+1$ with c_p parts of length p , the signed multiplicity of the kernel labelled by λ at uniform coupling on K_n factorises into four combinatorial pieces, each derivable from the GF:

$$\mu_{\lambda}^{(n)}(m) = \underbrace{\frac{k!}{\prod_{p \geq 1} c_p!}}_{\text{time-ordering}} \times \underbrace{A_{00}^{k-1}(K_n)}_{\text{rail-labeling}} \times \underbrace{(n-1)^{m+1-k}}_{\text{stays}} \times \underbrace{(-1)^{k-1}}_{\text{swap sign}} \quad (7)$$

where $A = J_n - I_n$ is the no-self-loop adjacency on K_n and $A_{00}^{k-1}(K_n) = [(n-1)^{k-1} + (n-1)(-1)^{k-1}]/n$ is the number of length- $(k-1)$ rail-jump sequences $0 \rightarrow 0$. *Each factor is the answer to one combinatorial question*; together they give the integer coefficient of the kernel in $W_m^{(n)}$.

Sanity-check at $n = 3, m = 5$ (verified in `verify_symmetry_factor.py`). The 8 chunk patterns and their multiplicities:

λ	k	time-ord.	A_{00}^{k-1}	$(n-1)^{m+1-k}$	μ
(6)	1	1	1	32	+32
(4, 1, 1)	3	3	2	8	+48
(3, 2, 1)	3	6	2	8	+96
(3, 1, 1, 1)	4	4	2	4	-32
(2, 2, 2)	3	1	2	8	+16
(2, 2, 1, 1)	4	6	2	4	-48
(2, 1, 1, 1, 1)	5	5	6	2	+60
(1, 1, 1, 1, 1, 1)	6	1	10	1	-10
sum = $\frac{2}{3} \cdot 3^5 =$					+162

The signs come from $(-1)^{k-1}$ (parity of swaps), the rest are non-negative integers. The same formula reproduces $n=3, m=4$ and $n=4, m=4$ exhaustively (`verify_symmetry_factor.py`).

What this means for Gunnar’s “the symmetry factors are taken care of”. The Laplacian formula bundles the $k!/\prod c_p!$ time-ordering, the A_{00}^{k-1} rail-labeling, the $(n-1)^{m+1-k}$ partner-choice, and the $(-1)^{k-1}$ swap-sign into a single matrix element $(L^m)_{00}$. Equation (7) *unbundles* them on demand: if you want the integer that multiplies a specific kernel, you read off four numbers from the partition itself (no diagrammatic enumeration). This is the AC analogue of the standard “count the automorphism group” calculation in Feynman diagrammatics — but expressed as a formula, not an algorithm.

4 Shining a light: marking, the resolvent, and Schur reduction

The Laplacian collapses information by design — but exactly how much it collapses is a knob the user controls, by choosing what to label and what to leave uniform. This section gathers three algebraic constructions, all within scope of Flajolet & Sedgewick’s machinery for walks on weighted graphs (*Analytic Combinatorics*, ch. V), that turn the rail-walk picture into something more flexible than “one number per m .”

(1) The walk generating function and its closed form (F-S Theorem V.7). The single most useful algebraic move is the resolvent. Flajolet & Sedgewick’s Theorem V.7 (*Asymptotics of paths in graphs*, §V.5, p. 343) states that for any non-negative weighted adjacency matrix T the matrix $F(z) = (I - zT)^{-1}$ has all entries with common radius of convergence $\rho = \lambda_1^{-1}$ (largest eigenvalue) given by the smallest positive root of $\det(I - zT) = 0$. Specialising to our L ,

$$\mathcal{G}^{(n)}(z) := \sum_{m \geq 0} W_m^{(n)} z^m = [(I - zL)^{-1}]_{00} = \frac{\det(I - zL)_{[0,0]}}{\det(I - zL)}, \quad (8)$$

with the numerator being the cofactor obtained by deleting row/column 0. For uniform coupling $v_{ij}=v$ on K_n , the spectrum $\{0, nv, nv, \dots\}$ gives

$$\boxed{\mathcal{G}^{(n)}(z) = \frac{1 - zv}{1 - znv}} \quad (9)$$

which expands to $1 + \sum_{m \geq 1} \frac{n-1}{n} (nv)^m z^m$, recovering (1) as the coefficients of a rational function. The pole at $z = 1/(nv)$ is the asymptotic-growth radius; the zero at $z = 1/v$ is what suppresses the all-uniform mode.

Example (1') Resolvent at $n=3$ uniform

Specialise (9) to $n=3$: $\mathcal{G}^{(3)}(z) = \frac{1 - zv}{1 - 3zv}$. SymPy expansion (`verify_shining.py`, test 1):

$$\mathcal{G}^{(3)}(z) = 1 + 2vz + 6v^2z^2 + 18v^3z^3 + 54v^4z^4 + \dots$$

i.e. coefficients $\frac{2}{3}3^m$ for $m \geq 1$ as predicted. The pole at $z = 1/(3v)$ is the Perron eigenvalue $\lambda_1=3v$; the zero at $z=1/v$ trims the prefactor from 3^m to $\frac{2}{3}3^m$.

(2) First-return decomposition. The standard “hitting-time” factorisation $\mathcal{G}(z) = 1/(1 - F(z))$ — where $F(z)$ is the GF of walks that return to rail 0 *for the first time* — is implicit in the resolvent identity $F = I + zTF$ used in the proof of Theorem V.7 (equation (85) of F-S, p. 343), and predates F-S as standard renewal theory. Inverting (9):

$$F^{(n)}(z) = 1 - \frac{1}{\mathcal{G}^{(n)}(z)} = \frac{(n-1)zv}{1 - zv}.$$

The simple pole at $z = 1/v$ is the “intrinsic” single-rail cost; the factor $(n-1)$ counts the off-rail destinations. This separates the *single-rail* structure (the $1/(1 - zv)$ factor on a fixed rail) from the *branching factor* $(n-1)$ at each vertex of the walk — the cleanest decomposition of why the answer scales as $\frac{n-1}{n} (nv)^m$ at all.

Example (2') First-return GF at $n=3$

With $n=3$, $F^{(3)}(z) = \frac{(n-1)zv}{1 - zv} \Big|_{n=3} = \frac{2zv}{1 - zv}$. Expansion (`verify_shining.py`, test 2):

$$F^{(3)}(z) = 2vz + 2v^2z^2 + 2v^3z^3 + 2v^4z^4 + \dots$$

Reading: *exactly two* first-return walks at every length $m \geq 1$ in the unit-coupling limit — the two off-rail destinations $\{1, 2\}$. The $1/(1 - zv)$ factor exposes the freedom to make any number of identity-preserving stays between the leave and the return.

(3) Marking individual couplings (multivariate AC). Keep some subset of the v_{ij} symbolic and set the rest equal to v . The coefficient of any monomial $\prod v_{ij}^{p_{ij}}$ in $\mathcal{G}^{(n)}(z) = [(I - zL)^{-1}]_{00}$ counts walks weighted by the *exact number of times each coupling is used*. This is the F-S multivariate-marking construction (*Analytic Combinatorics*, §III.1 “An introduction to bivariate generating functions”, p. 152, with Theorem III.1 “Inherited parameters and ordinary MGFs”, p. 166): each v_{ij} is a marker variable for the corresponding atom (a coupling event between rails i and j). Practical uses:

- **Channel sensitivity.** Set all $v_{ij} = v$ except v_{01} ; then $\partial_{v_{01}}^k \mathcal{G}^{(n)}(z)|_{v_{01}=v}$ counts walks containing exactly k visits through the 0–1 coupling. The dominant channel for a chosen response is read off directly.
- **Sub-rail focus.** Distinguish a “foreground” rail subset $I \subset \{0, \dots, n-1\}$ from a “background” subset B . Keeping $\{v_{ij}\}_{i,j \in I}$ symbolic and lumping background couplings $\{v_{ij}\}_{i \in I, j \in B} = v$ produces a polynomial in the foreground couplings whose coefficients encode how the background contributes order-by-order.
- **Forbidden patterns.** Setting $v_{ij}=0$ deletes the i – j edge from the rail graph, restricting to walks that never use that channel. The same Laplacian formula then runs on the smaller graph; the systematic treatment of walks-with- constraints is the kernel method, F-S §VII.8.1 “Walks and the kernel method”, p. 506.

The point is that *the Laplacian is a multivariate polynomial in the v_{ij} , not a scalar*; treating it as such turns the closed form $(L^m)_{00}$ into a complete bookkeeping device for any per-channel question.

Example (3') Channel sensitivity & sub-rail focus at $n=3, m=2$

Start from the symbolic case ($n=3, m=2$): $(L^2)_{00} = 2v_{01}^2 + 2v_{01}v_{02} + 2v_{02}^2$ (verified in `poc_laplacian.py`, test 2).

Mark v_{01} , lump rest as v ($v_{02}=v_{12}=v$). Substitute and collect:

$$(L^2)_{00}|_{\text{marked}} = \underbrace{2v^2}_{[v_{01}^0]} + \underbrace{2v v_{01}}_{[v_{01}^1]} + \underbrace{2v_{01}^2}_{[v_{01}^2]}.$$

Reading: the six length-2 walks split as 2 with no 0–1 visits, 2 with exactly one, 2 with two.

Foreground $\{0,1\}$, *background* $\{2\}$. Same substitution, re-read by physical role: $2v_{01}^2$ (foreground only) + $2v_{01}v$ (mixed) + $2v^2$ (pure background). **Reading:** half the order-2 weight is “foreground only,” a quarter is “one foray into the background,” a quarter is “pure background dressing.” (Both readings verified in `verify_shining.py`, tests 3–4.)

Example (3'') Forbidden patterns (F-S §VII.8.1) at $n=3, m=3$

Set $v_{12}=0$ (delete the 1–2 edge: hub graph through rail 0 only). At $m=3$ the symbolic Laplacian gives

$$(L^3)_{00}^{\text{full}} - (L^3)_{00}^{v_{12}=0} = v_{12}(v_{01} - v_{02})^2.$$

Reading: the only $m=3$ contribution requiring the 1–2 edge is the symmetric interference between the two “via rail 0” channels — a clean “frustration” signature in the Kondo sense (verified in `verify_shining.py`, test 5).

(4) Schur reduction: integrating out a subset of rails. The most powerful “shine a light on I , marginalise out B ” construction is the Schur complement on the resolvent. Partition rails

into I (foreground) and B (background), and write

$$L = \begin{pmatrix} L_{II} & L_{IB} \\ L_{BI} & L_{BB} \end{pmatrix}, \quad (I - zL)^{-1} = \begin{pmatrix} \cdot & \cdot \\ \cdot & \cdot \end{pmatrix}.$$

Block-inversion gives the foreground sector

$$[(I - zL)^{-1}]_{II} = \left(I_{|I|} - zL_{II} - z^2 L_{IB} (I - zL_{BB})^{-1} L_{BI} \right)^{-1}.$$

Read this as: the foreground rails see a *renormalised effective Laplacian*

$$L_{II}^{\text{eff}}(z) = L_{II} + z L_{IB} (I - zL_{BB})^{-1} L_{BI},$$

which is the z -dependent self-energy contribution from background-rail excursions. This is the AC analogue of integrating out a fast field, and it *exposes* the foreground structure (rail labels still distinct in I) while *collapsing* the background (single z -dependent correction). The poles of L^{eff} are background eigenvalues inherited as renormalisation thresholds. Compare to the multi-channel Dyson equation (§5(B)): same algebraic move, different physical labels.

Example (4') Schur reduction at $n=3$ uniform: integrate out rails $\{1, 2\}$

With $L_{II}=2v$, $L_{IB}=(-v, -v)$, $L_{BI}=L_{IB}^\top$, $L_{BB}=v \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, plug into the boxed formula:

$$\Sigma(z) := z L_{IB} (I_2 - zL_{BB})^{-1} L_{BI} = \frac{2zv^2}{1 - zv}, \quad L_{00}^{\text{eff}}(z) = 2v + \Sigma(z) = \frac{2v}{1 - zv}.$$

Then

$$(1 - z L_{00}^{\text{eff}})^{-1} = \frac{1 - zv}{1 - 3zv} = \mathcal{G}^{(3)}(z), \quad \checkmark$$

exactly the closed form of (1'). **Reading:** integrating out rails $\{1, 2\}$ produces a frequency-dependent self-energy $\Sigma(z)$ on rail 0 that exactly reproduces the full answer; the pole structure of Σ encodes which background eigenmodes have been folded in. Verified end-to-end in `verify_shining.py`, test 6, including Σ and L^{eff} separately.

(5) Continued fractions for walks with constraints (F-S Proposition V.3, p. 322).

A subtler F-S trick is the Jacobi continued-fraction representation of weighted walks on a one-dimensional support (§V.4 “Nested sequences, lattice paths, and continued fractions”, p. 318). For walks on $\mathbb{Z}$ with self-loop weights b_k and edge weights a_k ,

$$\sum_m [(\text{walks } 0 \rightarrow 0 \text{ of length } m)] z^m = \frac{1}{1 - b_0 z - \frac{a_1^2 z^2}{1 - b_1 z - \frac{a_2^2 z^2}{1 - b_2 z - \dots}}}.$$

On a general graph this generalises to a *matrix* continued fraction obtained by recursive Schur reduction on a chosen sequence of rails. For our K_n with the 1 on rail 0, peeling rails one at a time from B gives a length- $(n-1)$ continued fraction whose levels correspond to “walks that have left rail 0 exactly k times.” Useful when one wants the contribution *cut at level k* (i.e. walks that use at most k off-rail excursions).

Example (5') Matrix continued-fraction peel at $n=3$

The Schur reduction in (4') is precisely the depth-1 truncation of the matrix continued fraction obtained by peeling off-rails one at a time. The *depth* is the number of off-rail excursions retained: depth 0 returns the bare propagator, depth 1 keeps single off-rail visits, depth $\geq n-1$ recovers the full $\mathcal{G}^{(n)}(z)$ of example (1'). For $n=3$ uniform, peeling rail 2 from L^{eff} of (4') collapses the bottom-most level to the scalar $1/(1-zv)$, giving the depth-2 Jacobi-style continued fraction

$$\mathcal{G}^{(3)}(z) = \frac{1}{1 - \frac{2zv}{1-zv}} = \frac{1-zv}{1-3zv},$$

the rail-graph analogue of F-S Proposition V.3 with depth controlled by the number of allowed excursions instead of Motzkin height.

What this buys, in one paragraph

The Laplacian with symbolic v_{ij} is a polynomial generating function in $|E(K_n)|$ variables; its $(0,0)$ -resolvent (8) packages *all* order- m rail-walk counts simultaneously. Setting some $v_{ij} = v$ collapses uninteresting channels (losing their internal structure) while keeping others symbolic *exposes* the channels you care about. Schur reduction on a sub-graph integrates the rest into a frequency-dependent self-energy without losing the foreground detail. In Flajolet–Sedgewick language, all of these are textbook moves on rational walk generating functions. The diagrammatic “splendour” Gunnar wants to keep visible is controlled exactly by which v_{ij} you choose to leave symbolic.

5 What AC reveals — and what it deliberately hides

Picking up your earlier remark that the AC formalism risks burying “the full splendour of the diagrammatics”: here is a clean accounting of what is visible vs. hidden in the Laplacian/walk picture, for this specific problem.

Visible / made manifest	Hidden / collapsed by symmetry
Eigenvalue structure of the rail graph: L 's spectrum determines the asymptotic growth $\sim (nv)^m$ for any n .	The cyclic order in which a fixed rail is visited; only the visit count p survives.
The dominant rail mode (largest λ_α) and its eigenvector — i.e. which couplings drive the leading order in inhomogeneous v_{ij} .	Distinction between bridges and swaps at uniform coupling: the Laplacian combines them additively.
Asymptotics for free: $W_m \sim (\text{leading } \lambda)^m$, and corrections come from the gap.	The full “shape” of a single diagram (whether it's a long stay-chain broken by short swaps or vice versa); only the topological invariant matters.
Loop kernels factor multiplicatively against the path weight.	The way crossings in the time order are resolved (this becomes the visit-pattern label, which is a partition — much coarser than the original diagram).

When the splendour matters: if one asks about the *distribution of visit patterns* (e.g. at fixed m , what fraction of the weight sits in the all-tadpole sector vs. the long-bubble sector?), the

Laplacian collapse is too coarse and the original diagrammatics is the right object. If one asks for the *leading-m scaling* or for *symbolic identities* between v_{ij} , the Laplacian sees it directly. This is the same trade-off as in your DP comment: AC compresses “symmetry-factor accounting + sign tracking + integral library $\rightarrow$ spectrum + finite kernel set,” and the price is that any question *above* the Laplacian (rail-by-rail observables, individual diagram shapes) requires going back to the raw enumeration.

6 Verification (numerical PoC)

We brute-force-enumerate the rail walks for small (n, m) and check against (1) and the symbolic L -formula. Two PoC scripts accompany this note.

`poc_laplacian.py` (base verification):

- For uniform $v = 1$, $n \in \{2, 3, 4, 5\}$, $m \in \{0, \dots, 8\}$: direct enumeration of all signed rail walks $0 \rightarrow 0$ of length m , compared against $\frac{n-1}{n}n^m$. Match to machine precision.
- For symbolic v_{ij} , $n = 3$, $m \in \{1, \dots, 4\}$: enumeration into a polynomial in $\{v_{01}, v_{02}, v_{12}\}$, compared against $(L^m)_{00}$ computed via SymPy. Identical polynomials.
- Chunk-length pattern decomposition at $n=3$, $m=5$: 8 distinct chunk-patterns appear (each one a partition of $m+1=6$, bounded above by $p(6)=11$), with signed multiplicities summing to $\frac{2}{3} \cdot 3^5 = 162$.

`verify_shining.py` (every §4 example):

- (1') resolvent $\mathcal{G}^{(3)}(z) = (1 - zv)/(1 - 3zv)$ via SymPy, matching the spectral-decomposition prediction;
- (2') first-return $F^{(3)}(z) = 2zv/(1 - zv)$;
- (3') channel-sensitivity polynomial $2v^2 + 2v v_{01} + 2v_{01}^2$ at $m=2$;
- (4') sub-rail-focus decomposition (foreground / mixed / pure-background);
- (3'') forbidden $v_{12}=0$ at $m=3$, with the killed terms factoring as $v_{12}(v_{01} - v_{02})^2$;
- (4') Schur reduction including intermediate $\Sigma(z) = 2zv^2/(1 - zv)$ and $L_{00}^{\text{eff}}(z) = 2v/(1 - zv)$.

All checks pass; output reproduces the formulae quoted in §4 boxes.

7 Literature: what game is this, really?

The first thing to ask is not “where does the Laplacian appear in the literature” (it is everywhere) but “*what physical problem class is your note actually a member of?*” The setup is recognisable:

- one distinguished particle (the 1, on rail 0) propagating through a bath of $n-1$ species (the 0-rails);
- two vertex types coupling the 1 to one bath species at a time: $+v_{ij}$ when the 1 *keeps* its species (direct channel), $-v_{ij}$ when it *swaps* with a bath particle (exchange channel);
- bath species enter symmetrically (no preferred bath rail);
- every diagram is enumerated by “the path the 1 takes.”

That is, almost word for word, the standard description of a **single-impurity / mobile-impurity / multi-channel problem with direct + exchange scattering channels**. The physical homes for this game are well-mapped:

(A) Multi-channel Kondo and the impurity star graph. A single impurity spin (the 1) coupled to k conduction-electron flavour channels (the $n-1$ bath rails) with antiferromagnetic exchange is the multi-channel Kondo problem of Nozières & Blandin (1980). The *skeletal* object that controls its physics is the **star graph**: one central node (the impurity) connected to k outer nodes (the channels). Recent work (Patra & Lal, “Frustration shapes multi-channel Kondo physics: a star graph perspective,” arXiv:2205.00790, *J. Phys. Condens. Matter* 35 (2023)) makes this absolutely explicit: the multi-channel Kondo skeletal graph *is* the star $K_{1,n-1}$, and the bookkeeping operator is its graph Laplacian. Our K_n with the 1 on rail 0 is the same star combinatorially (in your set-up the 0-rails can also bridge to one another, so the graph is upgraded from a star to the full K_n , but the distinguished-vertex structure is identical).

- Nozières & Blandin, “Kondo effect in real metals,” *J. Phys.* 41 (1980), 193 (origin of multi-channel Kondo).
- Andrei & Destri, “Solution of the multichannel Kondo problem,” *Phys. Rev. Lett.* 52 (1984), 364.
- Affleck & Ludwig, “Critical theory of overscreened anisotropic Kondo,” *Nucl. Phys. B* 352 (1991), 849 (channel index = species index).
- Patra & Lal, “Frustration shapes multi-channel Kondo physics: a star graph perspective,” arXiv:2205.00790.

(B) Mobile-impurity / polaron diagrammatic Monte Carlo. When the 1 is a heavy or mobile particle and the bath is a continuum of phonons, bosons, or fermions, the same diagram class — a single impurity worldline interacting with bath modes via vertex couplings — is the **Fröhlich / Bose / Fermi polaron problem**. The diagrammatic Monte Carlo programme of Prokof’ev & Svistunov samples exactly the “1-walks-through-vertices” diagrams with sign-cancellation between the direct and exchange channels — the same $\pm v_{ij}$ structure you wrote down. “Bold” DiagMC re-sums these into a renormalised propagator on the impurity rail.

- Prokof’ev & Svistunov, “Bold diagrammatic Monte Carlo: a generic sign-problem-tolerant technique for polaron models,” *Phys. Rev. B* 77 (2008), 125101; arXiv:0801.0911.
- Vlietinck et al., “Diagrammatic Monte Carlo study of a mass-imbalanced Fermi-polaron system,” *Phys. Rev. B* 91 (2015), 144507; arXiv:1411.1562.
- Mishchenko & Nagaosa, “Diagrammatic Monte Carlo method for many-polaron problems,” *Phys. Rev. Lett.* 113 (2014), 166402.
- Houtput & Tempere et al., first-principles polaron DiagMC, *Nature Phys.* (2025) (modern multi-component bath version).

The closed form $\frac{n-1}{n}(nv)^m$ is, in DiagMC language, the exact m -th-order contribution to the impurity self-energy at uniform coupling on K_n — and the sign-tolerance result of bold DiagMC is exactly the statement that signed sums of $(\pm v)$ vertices collapse onto a single controlled scalar (the eigenvalue nv of the Laplacian).

(C) Bethe–Salpeter / T-matrix ladders with direct and exchange. At the diagrammatic level, the rail-walk picture is the **Bethe–Salpeter ladder approximation in the impurity-bath two-body channel**, including both direct (Hartree, $+v$) and exchange (Fock, $-v$) rungs. Summing direct + exchange ladders at each rung is the standard particle-particle T-matrix expansion (Bethe–Goldstone). Each rail-walk of length m is one rung sequence; the Laplacian eigenvalue is the ladder-resummation pole.

- Bethe & Salpeter (1951), *Phys. Rev.* 84, 1232 (origin).
- Salpeter (2008 retrospective), arXiv:0811.1050.

- Romaniello, Bechstedt & Reining, *Phys. Rev. B* 85 (2012), 155131 (BSE with exchange and the T-matrix in multi-band systems).

The novel ingredient in your note relative to a textbook BSE is that the bath has $n-1$ *distinct* channels with arbitrary cross-couplings v_{ij} rather than a single bath; the Laplacian generalises the usual ladder-resummation eigenvalue to the spectrum of K_n .

(D) Anderson impurity / multi-orbital generalisations. For completeness: the same diagrammatic structure appears in the multi-orbital Anderson impurity model and its DMFT/CT-QMC solution, where “rails” are bath orbitals coupled to a localised impurity orbital with direct/exchange hybridisations.

- Anderson (1961), *Phys. Rev.* 124, 41 (origin of the model).
- Werner & Millis review, *Rev. Mod. Phys.* 83 (2011), 349 (CT-QMC sign tracking for multi-orbital impurity models).

This is the closest electronic-structure cousin of your set-up.

Summary of the game. Gunnar’s n -rail diagrams *are* the order-by-order perturbation series for a single impurity (the 1) propagating through an $n-1$ -species bath, with direct ($+v_{ij}$) and exchange ($-v_{ij}$) vertex channels — the multi-channel Kondo / mobile-polaron / BSE-ladder family. The skeletal object common to threads (A)–(D) is the impurity star graph; on it, the matrix element $(L^m)_{00}$ is the m -th order impurity-projected self-energy at the ladder-resummation level. The AC contribution proper (graph-Laplacian / Kirchhoff matrix-tree bookkeeping, Symanzik-polynomial loop machinery, Doi–Peliti vertex assembly, walk generating functions on weighted graphs) is the infrastructure that makes the closed form $\frac{n-1}{n}(nv)^m$ fall out cleanly — those pieces are part of the standing CFAC programme and are not re-litigated here.

8 Notes & caveats

We are confident in the combinatorial half of the answer (§2): the closed form $\frac{n-1}{n}(nv)^m$, and the matrix-element form $(L^m)_{00}$ for symbolic v_{ij} , is verified three ways (constructional, symbolic via SymPy, numerical brute-force enumeration up to $n=5$, $m=7$). The analytic half (§3) is a correct *framework* but an unfinished *calculation*: the kernels per chunk-pattern are sketched, not derived. Below is a list of the assumptions we made and the places where Gunnar’s intent could differ from ours.

1. **Diagram class.** We assume every vertex has the 1’s rail as one of its two participants (this is what your local rule “ $+v_{ij}$ for bridges the 1 does not take, $-v_{ij}$ for rail-changes” implies). If you also intend to admit *purely-0* vertices — two off-path rails meeting with no 1-line involved — the Laplacian undercounts those. We do not believe this is your intent, but flag it explicitly.
2. **Connectedness.** You wrote “connected only.” A rail-walk that visits every vertex is automatically connected, so we are consistent. If your notion of “connected” admits diagrams in which the 1-line itself branches (it does not, in a standard walk), the Laplacian would need extending to a higher-rank object.
3. **Loop integrals: framework + atlas, not full library.** §3 gives the factorisation (Laplacian path weight) $\times$ (kinematic kernel) and writes out the four canonical kernels (tadpole, bubble, swap-chain, sunset). The general kernel library is bounded by $p(m+1)$ chunk-patterns; we have not enumerated the full library at any given order. Natural follow-up: pick a small case (e.g. $n=3$, $m=3$) and produce all eight master integrals attached to its visit-patterns explicitly.

4. **Renormalisation is downstream.** Tadpole sub-diagrams are absorbed into mass renormalisation in any sensible scheme; we have not threaded counterterms through the Laplacian decomposition. The combinatorial sum is unaffected, but a physical observable will differ from $\Gamma^{(m)}$ by counterterm insertions.
5. **Splendour vs. collapse.** The Laplacian compresses sign, symmetry-factor, and rail-label into a single matrix — the win you asked about. As you noted earlier, this also *hides* information: the individual shape of a diagram, the cyclic order of bridges around a fixed off-path rail, and the way crossings are resolved are no longer visible. Any question that lives above the Laplacian (rail-by-rail observables, distribution of weight across diagram shapes) requires going back to the raw enumeration. § 4 (*Shining a light*) gives the algebraic dial that controls how much detail to keep, and § 5 makes the trade-off explicit.
6. **Sign convention.** A walk with k swaps carries $(-1)^k$. The sign of every term in $(L^m)_{00}$ then follows directly from the parity of swaps. Our PoC verifies this agrees with brute-force enumeration; if you are using a convention that absorbs an extra (-1) per closed 0-loop, our individual polynomial coefficients would shift in sign while $|W_m^{(n)}|$ is unchanged.
7. **Boundary projection vs. trace.** We compute $(L^m)_{00}$ — walks that start *and* end on rail 0. If you want walks that may end on *any* rail (sum over endpoints), replace the matrix element with $\mathbf{1}^\top L^m \mathbf{e}_0 = 0$ for $m \geq 1$ (because $\mathbf{1}$ is in the kernel of L); if you want a trace over all start/end rail pairs you get $\text{tr } L^m = \sum_{\alpha \geq 1} \lambda_\alpha^m$. We have used the source-and-sink-on-rail-0 convention, matching the diagrams you drew.

9 Summary

1. The transfer matrix for your “1-walks-through- n -rails” problem is the weighted graph Laplacian $L = D - A$ on K_n .
2. For uniform coupling, $W_m^{(n)} = \frac{n-1}{n}(nv)^m$. This is the clean extension of $\frac{1}{2}(2v_1)^m$.
3. For symbolic v_{ij} , $W_m^{(n)} = (L^m)_{00}$; spectral decomposition gives $\sum_\alpha \lambda_\alpha^m |\psi_\alpha(0)|^2$.
4. Loop integrals factor off the combinatorics: each diagram contributes (Laplacian path weight) $\times$ (kinematic kernel of its chunk-length pattern). The kernel library is finite, of size bounded by $p(m+1)$ at order m — a (very large) $\rightarrow$ (partition-number) contraction.
5. AC reveals the spectrum, asymptotics, and kernel library; it hides rail labels and time-ordering shape. Both sides of the trade-off are explicit.

Files in this letter: `note.tex/note.pdf` (this document), `poc_laplacian.py` (numerical/symbolic verification of $W_m^{(n)}$ and chunk-pattern decomposition), `verify_shining.py` (verification of every § 4 “shine a light” example), `README.md` (quick build instructions).